

2. (a) State the meaning of the term biological control. [1]

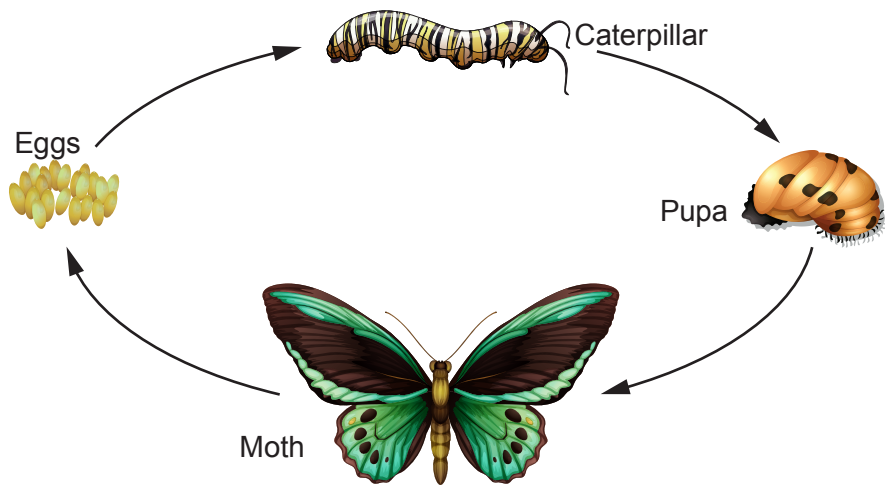
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(b) Many species of moths destroy food crops. The global effect of these pest species results in the loss of many billions of dollars per year. The stage in the life cycle of moths that causes the damage is the caterpillar, which eats the plant crop.

The simplified life cycle of most moths is shown in **Image 2.1**:

**Image 2.1**



**Table 2.2** shows five species of small wasps (*Trichogramma*) used as biological control agents. It also shows the stage in the moth life cycle they attack and the area of farmland in different countries where these biological control agents are already used successfully.

**Table 2.2**

| Species of wasp               | Stage of moth life cycle attacked by wasp | Food crops damaged by moth caterpillars | Region where wasp is used | Area of farmland where the wasp is used successfully (million hectares) |
|-------------------------------|-------------------------------------------|-----------------------------------------|---------------------------|-------------------------------------------------------------------------|
| <i>Trichogramma pretiosum</i> | eggs                                      | potatoes                                | China                     | 2.0                                                                     |
| <i>Trichogramma platneri</i>  | caterpillars                              | corn, sugar cane                        | Mexico                    | 1.5                                                                     |
| <i>Trichogramma acacioi</i>   | caterpillars                              | avocado and other fruit crops           | Brazil                    | 1.0                                                                     |
| <i>Trichogramma japonicum</i> | caterpillars                              | rice, potatoes                          | South East Asia           | 0.3                                                                     |
| <i>Trichogramma ostriniae</i> | caterpillars                              | corn                                    | Europe                    | 0.1                                                                     |



Examiner  
only

As an advisor for the United Nations you are asked to advise the Government of Peru on the use of a biological control agent to prevent moth damage in potato crops.

- (i) Using all the information given, suggest the best species of *Trichogramma* to use and give the reason for your choice. [2]

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- (ii) State **one** factor that would need to be investigated before *Trichogramma* is released into the environment in Peru. [1]

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- (iii) Suggest why it is not possible to conclude, from the information in the table, that the percentage of crops damaged by moths is less in China than it is in Europe. [1]

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